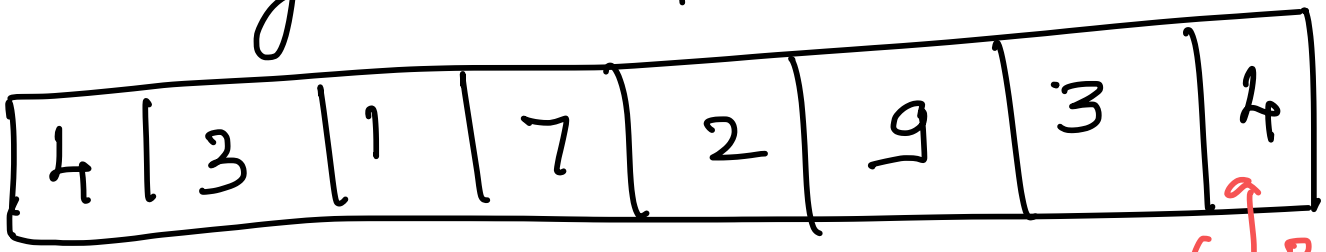
Miscellaneous Exercise Problems:

1. Implement all steps in
   (a) Bubblesort algorithm
   (b) selection sort algorithm
   (c) Merge sort algorithm
   (d) Quicksort algorithm to sort the following array

A = 

2. Prove that the bruteforce algorithm to find maximum in an array of n elements is asymptotically optimal.

3. Give the worst-case input for
   (i) Bubble sort
   (ii) Quick sort.

4. Illustrate Bucketsort for

$A =$

0.3	0.2	0.7	0.9	0.2	0.1	0.7	0.5	0.2	0.8

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5. Use Radix-Sort to sort the following list of binary numbers.

1011011011
1010110011
1000100010
1100110101
1010101011
1111111111
1000000001
1001100111001

Take octal base.

Mention why the intermediate sort must be stable!

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6. Compute $(17299271) * (20010102)$ using Karastuba's algorithm. How many multiplications have you used?

7. Implement Strassen's matrix multiplication for

$$\begin{bmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & 3 & 4 \\ 1 & 2 & 1 & 0 \\ 3 & 4 & 0 & 1 \end{bmatrix} * \begin{bmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & 3 & 4 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix}$$

8. Let X = binary digits of 65
Y = binary digits of 56.

Find $LCS(X, Y)$.

9. Implement matrix chain multiplication given

	order
A_1	6×5
A_2	5×4
A_3	4×3
A_4	3×2

to optimally parenthesize the product $A_1 A_2 A_3 A_4$. Verify your answer.

10. Give example of algorithm having exponential running time. Justify your answer.

11. Solve the following recurrences.

(a) $T(n) = 2T(n/2) + cn$

(b) $T(n) = 7T(n/2) + cn$

(c) $T(n) = T\left(\frac{9}{10}n\right) + T\left(\frac{1}{10}n\right) + cn$

(d) $T(n) = 2T(n-1) + 1$

Which algorithms give rise to above recurrences?

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12. Find the expected running time for Bucket sort. what is the Worst input for Bucket sort? Jushify your answer.

13. Give insertion sort algorithm. Analyze its Computational Complexity. Mention when it is used.

14. show that $\frac{n(n-1)}{2} \in \Theta(n^2)$ in two ways.